

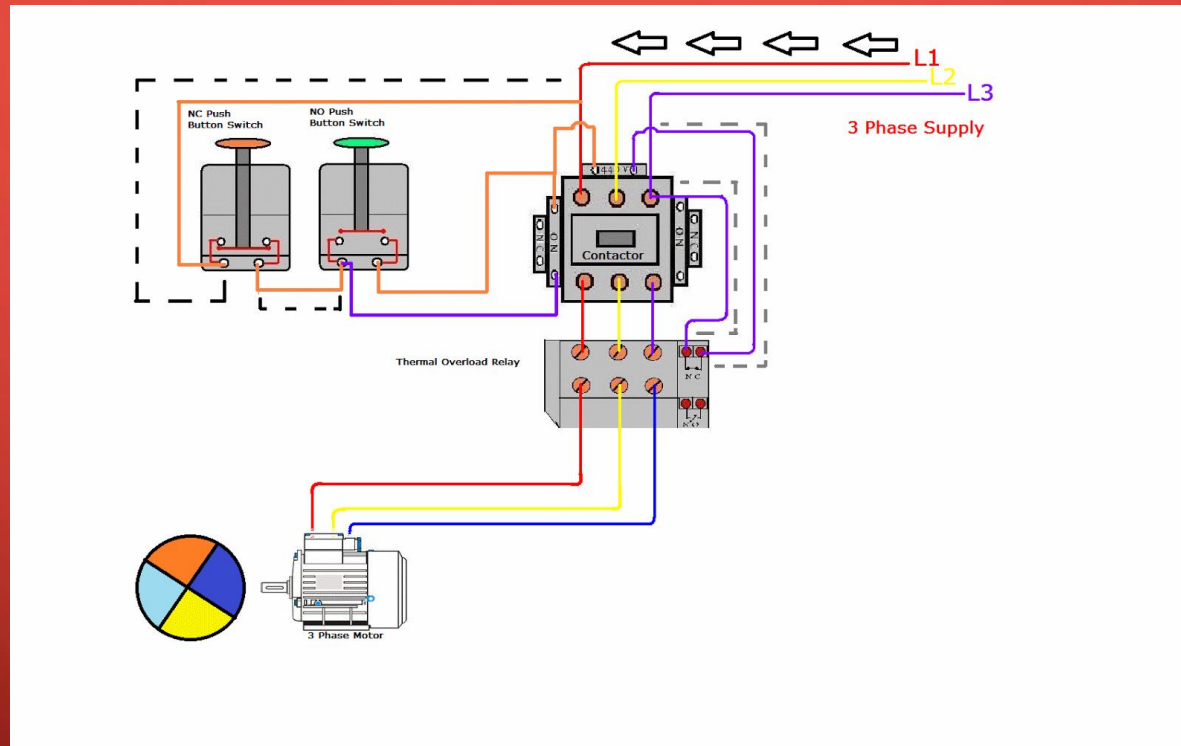
A decorative graphic on the left side of the slide, consisting of a network of orange lines and circles that resemble a circuit board or a neural network. The lines are of varying thickness and connect to small circles at various points, creating a complex, branching pattern that extends from the top to the bottom of the frame.

ELECTROMECHANICAL SYSTEM

CHAPTER 2: MOTOR CONTROL COMPONENTS

MOTOR CONTROL

A MOTOR CONTROL CIRCUIT CAN BE DEFINED AS A MEANS OF SUPPLYING POWER TO AND REMOVING POWER FROM A MOTOR.



BASIC MOTOR CONTROL COMPONENTS

1. CIRCUIT BREAKERS

2. FUSES

3. PUSH BUTTONS / SLIDE SWITCHES/ TOGGLE SWITCHES/ SELECTOR SWITCHES

4. INDICATOR LIGHTS

5. OVERLOAD RELAYS (THERMAL / SOLID STATE)

6. MAGNETIC CONTACTORS

7. CONTROL RELAYS

❖ **TIMERS**

❖ **COUNTERS**

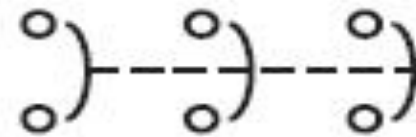
8. MOTOR STARTER

9. AC/DC MOTOR

10. VARIABLE FREQUENCY DRIVE (VFD)

1. CIRCUIT BREAKERS

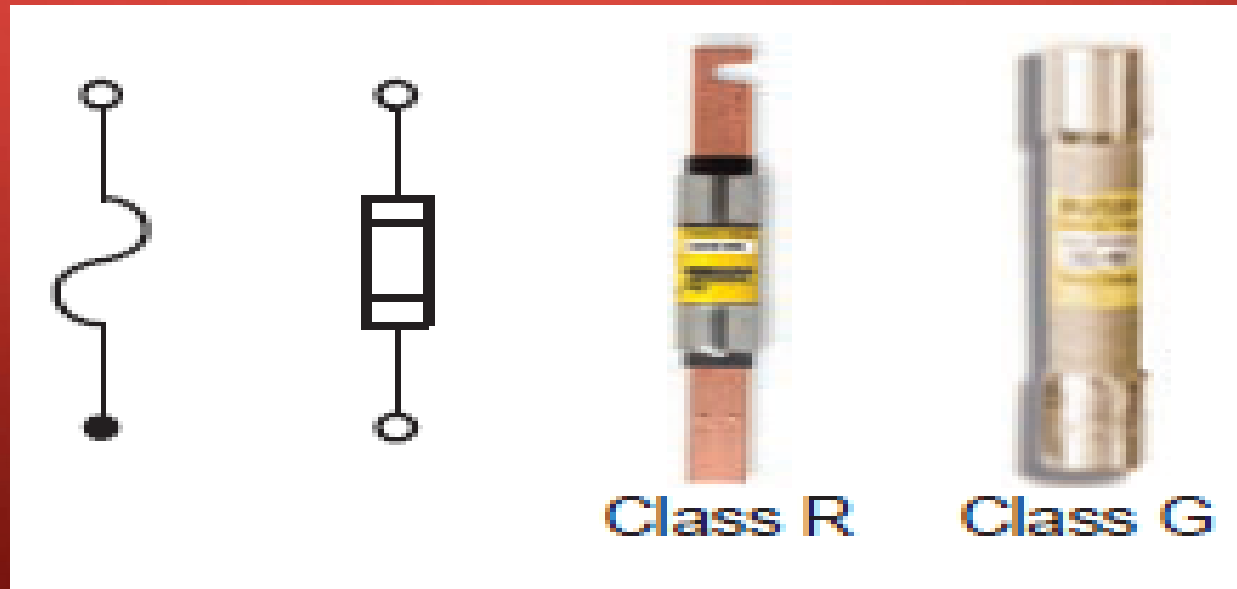
- A circuit breaker is a switching device that interrupts the abnormal or fault current. It is a mechanical device that disturbs the flow of high magnitude (fault) current and in additions performs the function of a switch.
- The circuit breaker is mainly designed for closing or opening of an electrical circuit, thus protects the electrical system from damage.



(b) Three-pole circuit breaker

2. FUSES

- AN ELECTRICAL SAFETY DEVICE THAT OPERATES TO PROVIDE OVERCURRENT PROTECTION OF AN ELECTRICAL CIRCUIT.
- ITS ESSENTIAL COMPONENT IS A METAL WIRE OR STRIP THAT MELTS WHEN TOO MUCH CURRENT FLOWS THROUGH IT, THEREBY STOPPING OR INTERRUPTING THE CURRENT.



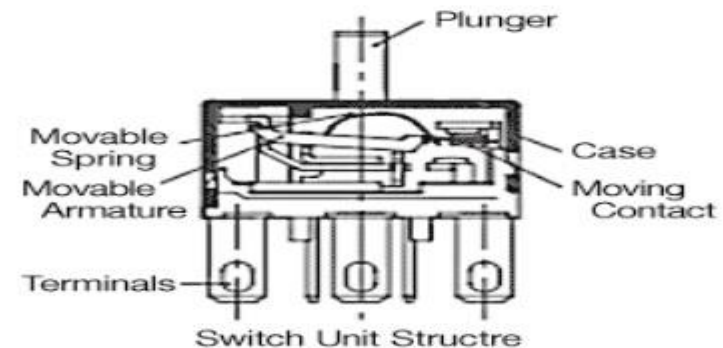
3. PUSH BUTTONS / SLIDE SWITCHES/ TOGGLE SWITCHES/ SELECTOR SWITCHES

- A CONTROL DEVICE USED TO MANUALLY OPEN AND CLOSE A SET OF CONTACTS.
- TYPICALLY USES OPERATING PLUNGER, A RETURN SPRING AND ONE SET OF CONTACTS.



Switch Unit

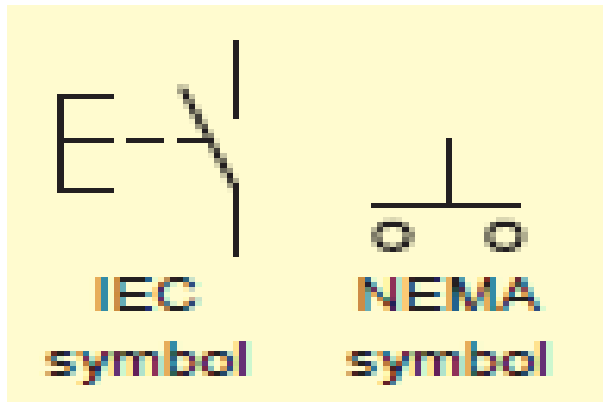
The external force or movement received by the operating part is relayed to the pushbutton inside the switch, and it moves the moving armature to open or close the contacts.



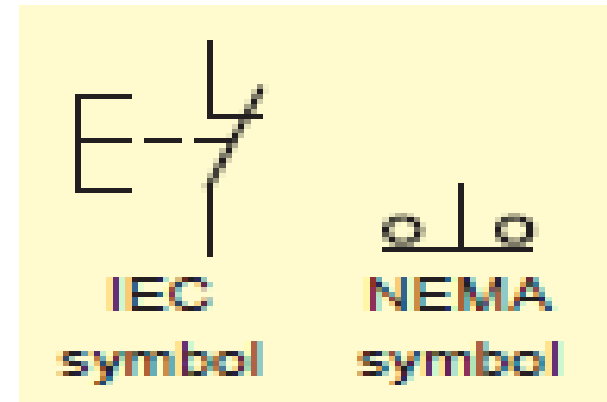
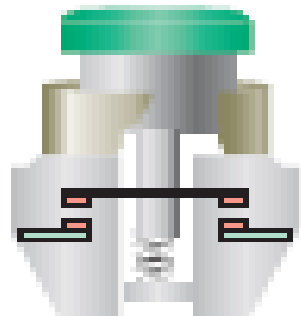
3. PUSH BUTTONS / TOGGLE SWITCHES/ SELECTOR SWITCHES

MOMMENTARY PUSH BUTTONS – THEIR CONTACTS REMAIN IN THEIR ACTIVATED STATE ONLY AS LONG AS THE BUTTON IS PRESSED

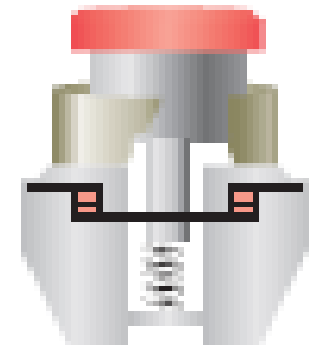
MAINTAINED CONTACT PUSH BUTTONS – WITH CONTACTS THAT REMAIN IN THEIR ACTIVATED STATE AFTER THE BUTTON IS RELEASED



N.O. (normally open)
pushbutton



N.C. (normally closed)
pushbutton



3. PUSH BUTTONS / TOGGLE SWITCHES/ SELECTOR SWITCHES

- A MANUALLY OPERATED SWITCH IS ONE THAT IS CONTROLLED BY HAND.
- A TOGGLE SWITCH USES A MECHANICAL LEVER MECHANISM TO IMPLEMENT A POSITIVE SNAP ACTION FOR SWITCHING OF ELECTRICAL CONTACTS. THIS TYPE OF SWITCHING OR CONTACT ARRANGEMENT IS SPECIFIED BY THE APPROPRIATE ABBREVIATION AS FOLLOWS:

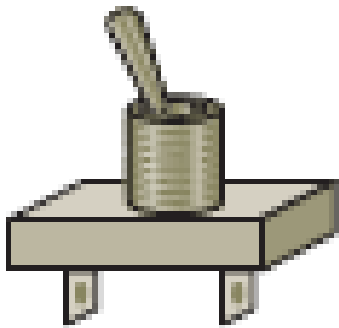
SPST—SINGLE POLE, SINGLE THROW

SPDT—SINGLE POLE, DOUBLE THROW

DPST—Double pole, single throw

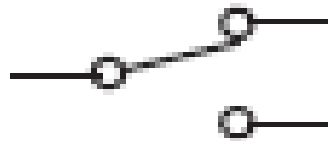
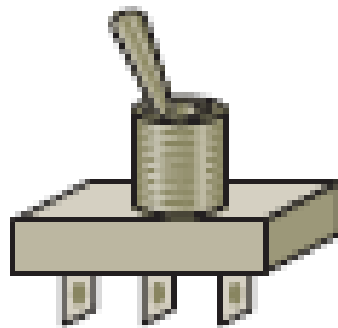
DPDT—Double pole, double throw

3. PUSH BUTTONS / SLIDE SWITCHES/ TOGGLE SWITCHES/ SELECTOR SWITCHES



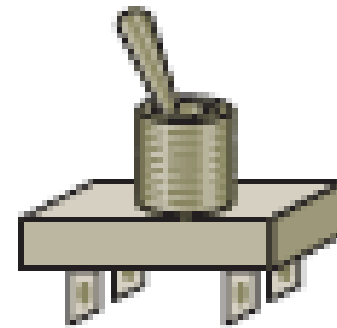
SPST

Single pole,
single throw



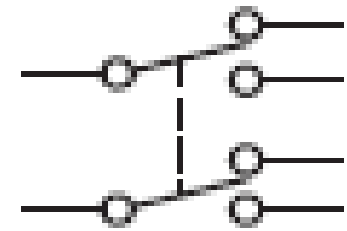
SPDT

Single pole,
double throw



DPST

Double pole,
single throw



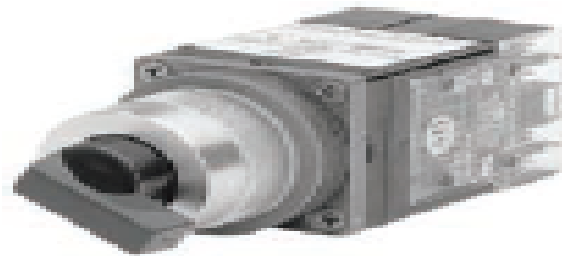
DPDT

Double pole,
double throw

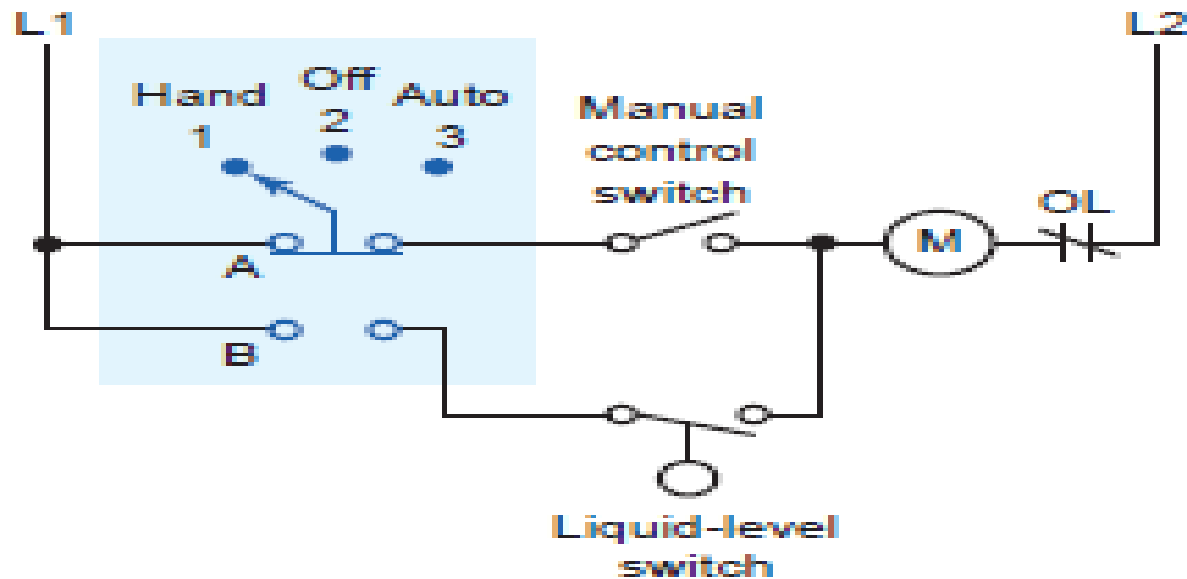
3. PUSH BUTTONS / TOGGLE SWITCHES/ SELECTOR SWITCHES

- ALSO USED TO MANUALLY OPEN AND CLOSE CONTACTS.
- THE DIFFERENCE BETWEEN A PUSH BUTTON AND SELECTOR SWITCH IS THE OPERATOR MECHANISM. A SELECTOR SWITCH OPERATOR IS ROTATED (INSTEAD OF PUSHED) TO OPEN AND CLOSE CONTACTS OF THE ATTACHED CONTACT BLOCK.
- AVAILABLE IN 2, 3, OR 4 POSITION TYPES.

3. PUSH BUTTONS / TOGGLE SWITCHES/ SELECTOR SWITCHES



Selector switch

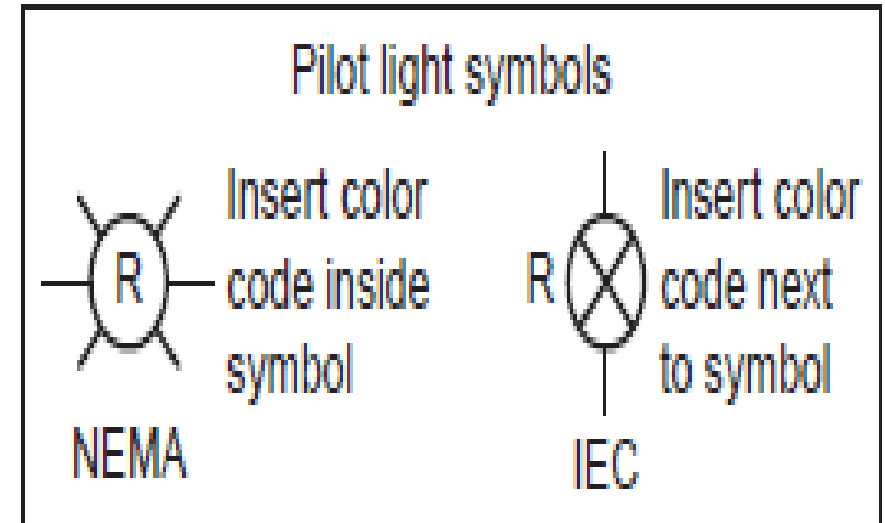


Position	Contacts	
	A	B
1	X	
2		
3		X

4. INDICATOR LIGHTS (PILOT LIGHTS)

- PROVIDE VISUAL INDICATION OF THE STATUS FOR MANY MOTOR-CONTROLLED PROCESSES PERMITTING PERSONNEL AT REMOTE LOCATIONS TO OBSERVE THE CURRENT STATE OF THE OPERATION.
- THEY ARE COMMONLY USED TO INDICATE WHETHER OR NOT A MOTOR IS OPERATING.

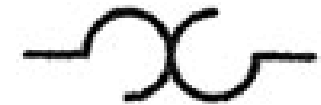
4. INDICATOR LIGHTS (PILOT LIGHTS)



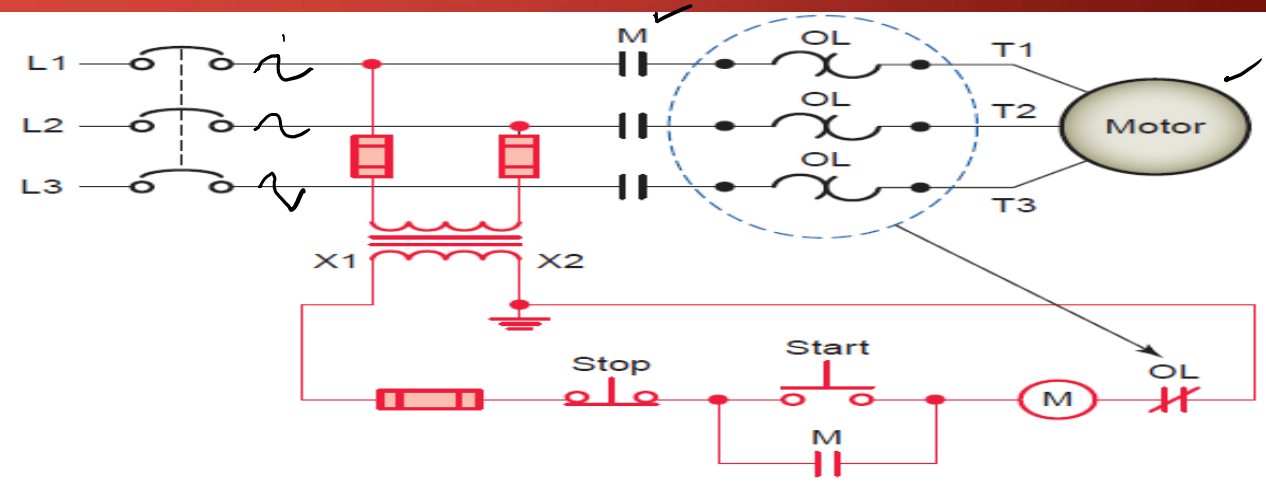
5. OVERLOAD RELAYS (THERMAL / SOLID STATE)

- USED TO PROTECT MOTORS FROM OVERHEATING.
- WHEN EXCESSIVE CURRENT IS DRAWN FOR A PREDETERMINED AMOUNT OF CURRENT, THE OVERLOAD RELAY'S CONTACTS OPEN, REMOVING POWER FROM THE MOTOR.
- AN OVERLOAD RELAY USED WITH A THREE PHASE MOTOR HAS 3 SUCH CONTACTS, ONE FOR EACH PHASE.

5. OVERLOAD RELAYS (THERMAL / SOLID STATE)



TR-N5/3

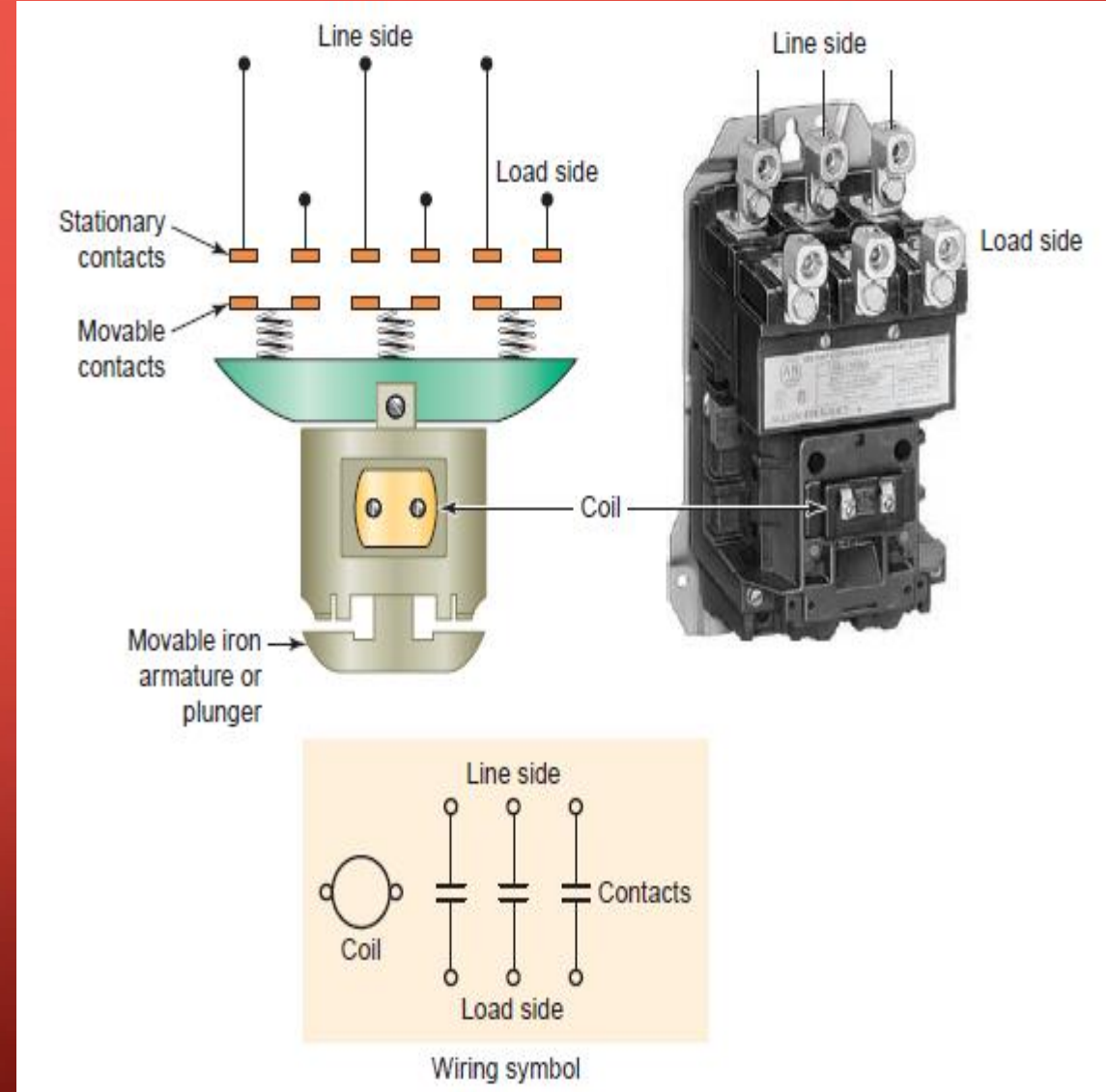


Thermal overload relay circuit operation

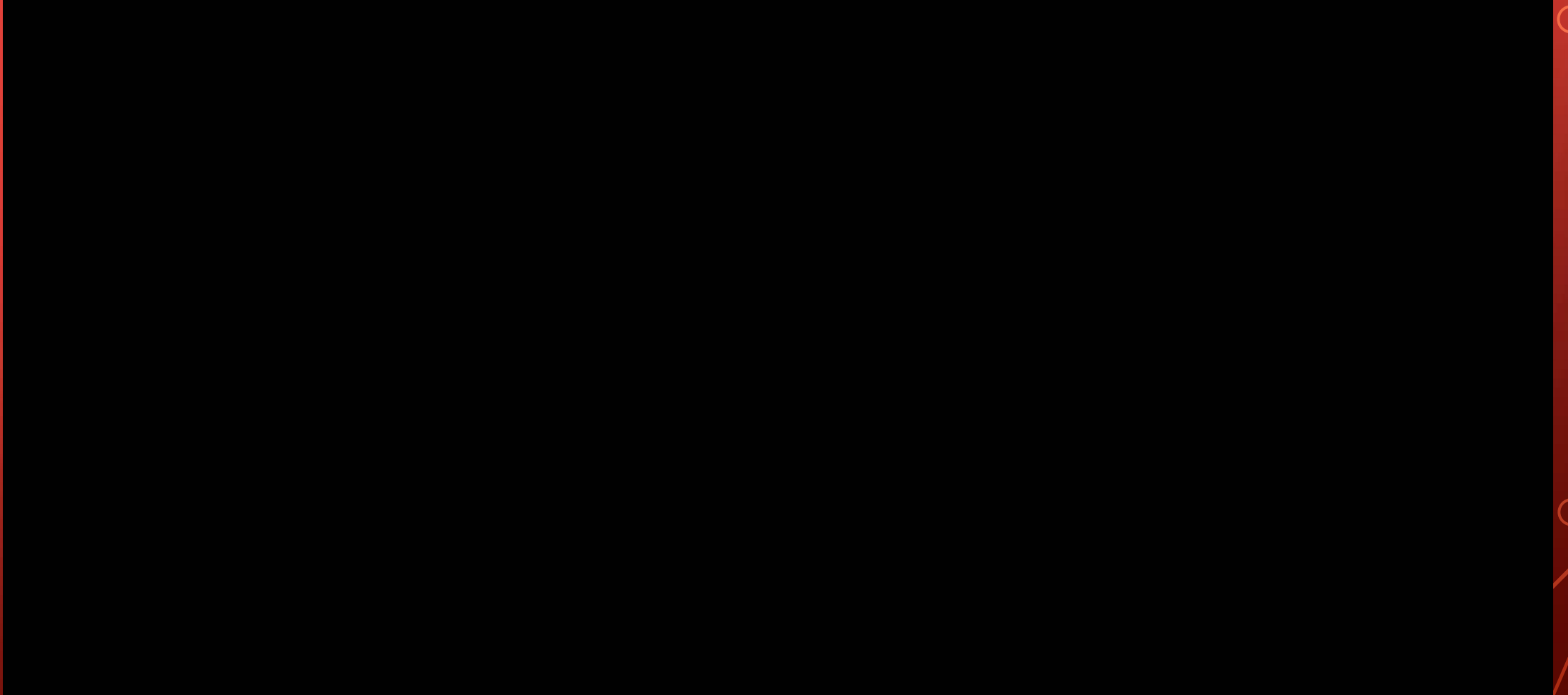
6. MAGNETIC CONTACTORS

- CONTACTORS ARE USED IN CONJUNCTION WITH PILOT DEVICES TO AUTOMATICALLY CONTROL HIGH-CURRENT LOADS.
- THE PILOT DEVICE, WITH LIMITED CURRENT HANDLING CAPACITY, IS USED TO CONTROL CURRENT TO THE CONTACTOR COIL, THE CONTACTS OF WHICH ARE USED TO SWITCH HEAVIER LOAD CURRENTS.
- THE BASIC PARTS ARE ELECTROMAGNETS, SPRING, COIL AND CONTACTS

6. MAGNETIC CONTACTORS



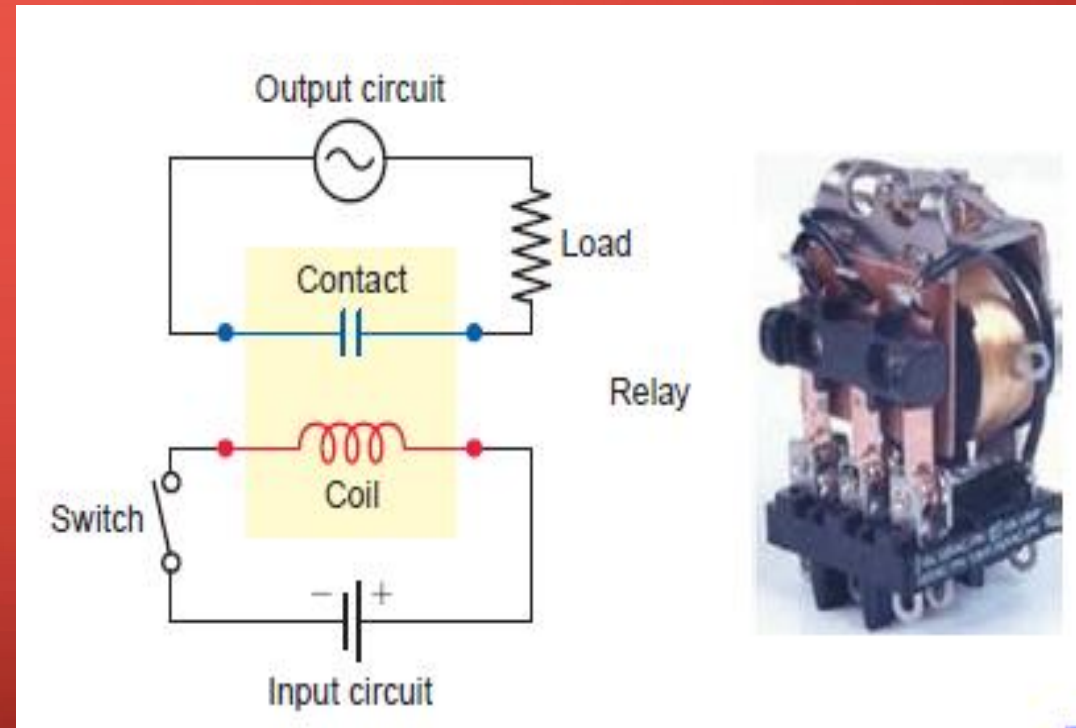
6. MAGNETIC CONTACTORS



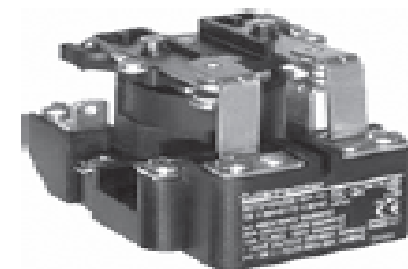
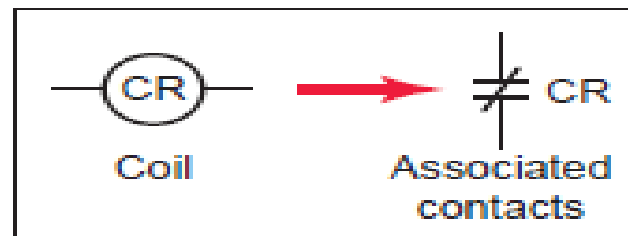
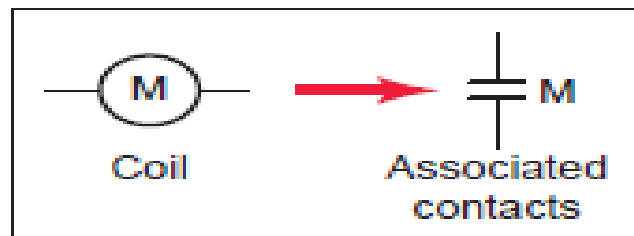
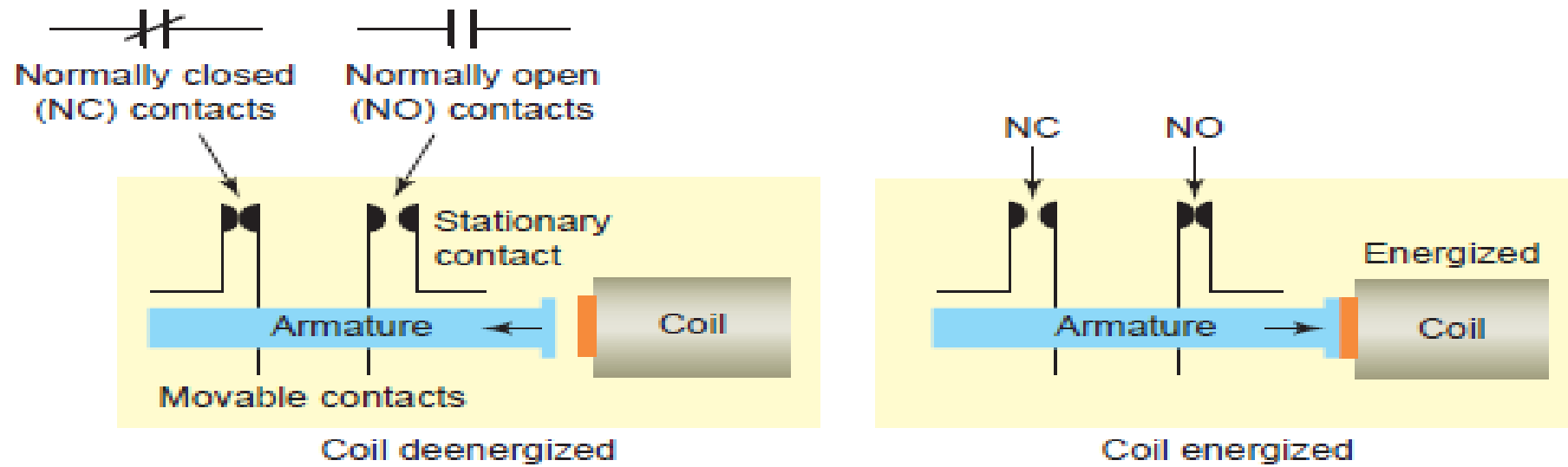
7. CONTROL RELAYS

- RELAYS ARE ELECTROMECHANICAL SWITCHES THAT ARE TRIGGERED BY SUPPLYING ELECTRICITY THROUGH ITS COIL. WHEN THE RELAY COIL IS ENERGIZED, ITS CONTACTS WILL SWITCH TO ITS OPPOSITE STATE.
- REMEMBER MAGNETIC CONTACTORS AND RELAYS ARE THE SAME.
- CONTACTORS ARE FOR HIGHER LOADS AND RELAYS ARE FOR LIGHTER LOADS.

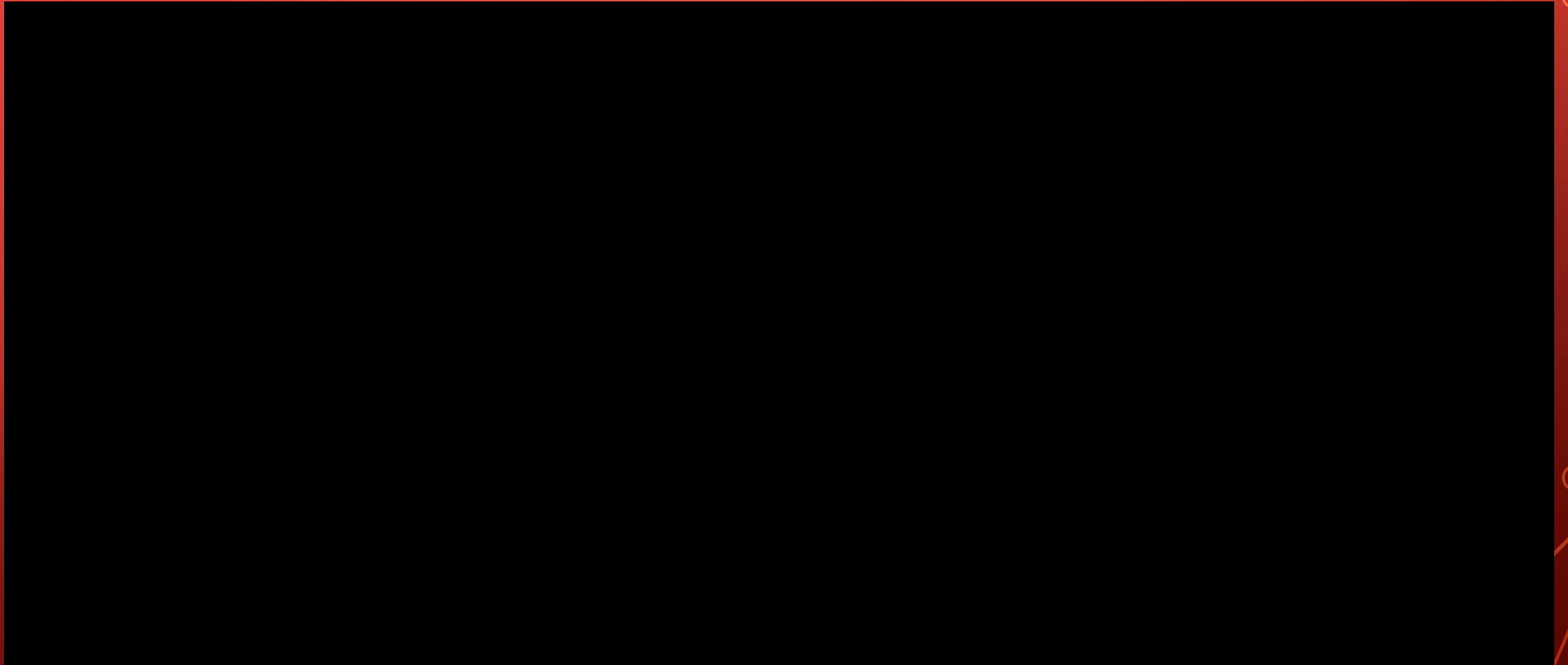
7. CONTROL RELAYS



7. CONTROL RELAYS



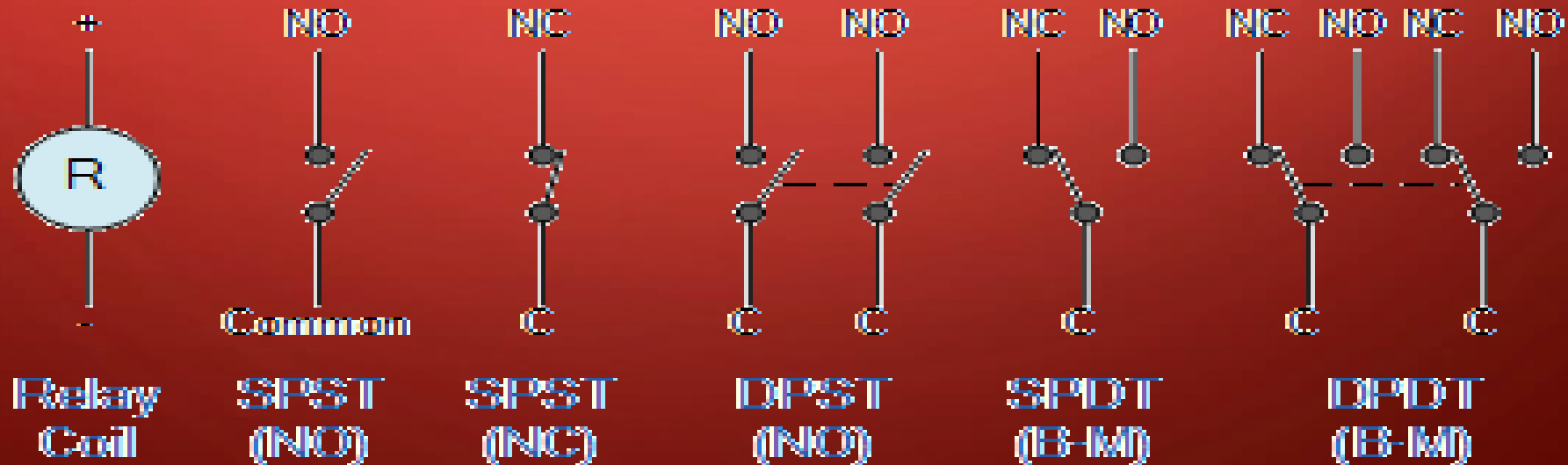
7. CONTROL RELAYS



7. CONTROL RELAYS

Electrical Relay Contact Configurations

- Where:
- C is the Common terminal
- NO is the Normally Open contact
- NC is the Normally Closed contact



7. CONTROL RELAYS

TIMER RELAYS – TYPE OF RELAY THAT DELAYS THE SWITCHING OF ITS CONTACTS FOR AN ADJUSTABLE SPECIFIED TIME.



7. CONTROL RELAYS

TIMER RELAYS

- CAN PERFORM EITHER ON-DELAY OR OFF-DELAY

ON-DELAY – INDICATES THAT A PRESET TIME MUST PASS AFTER THE TIMER RECEIVES A SIGNAL TO TURN ON BEFORE THE TIMER'S CONTACT CHANGE STATE.

OFF-DELAY – INDICATES THAT A PRESET TIME MUST PASS AFTER THE TIMER RECEIVES A SIGNAL TO TURN OFF BEFORE THE TIMER'S CONTACTS CHANGE STATE.

NC on-delay



NO on-delay



NO off-delay



NC off-delay



7. CONTROL RELAYS

COUNTERS – ELECTRO-MECHANICAL SWITCHES THAT ARE CAPABLE OF COUNTING AND STORING THE NUMBER OF OCCURENCES WHERE A PARTICULAR SIGNAL HAPPENS.

- COUNTER RELAYS HAS 2 COILS THE COUNTING COIL AND THE RESET COIL AND A CONTACT.

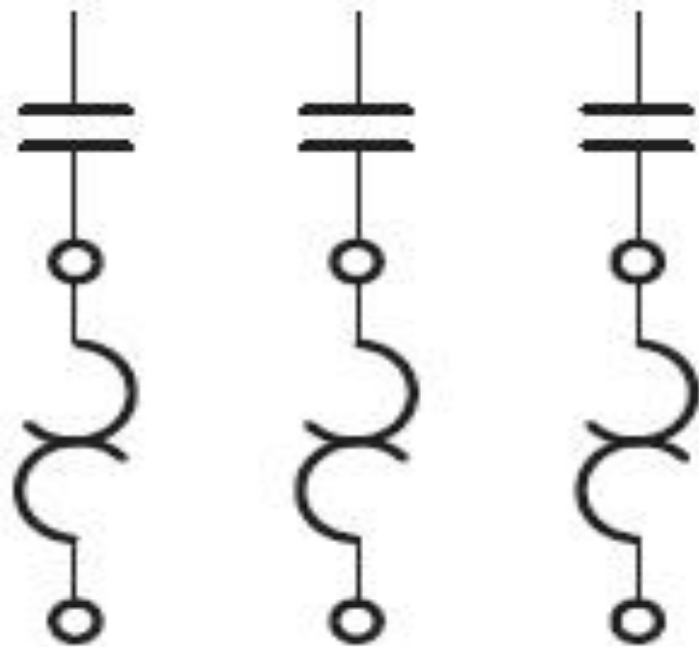


8. MOTOR STARTER

- WHEN A CONTACTOR IS COMBINED WITH AN OVERLOAD RELAY.

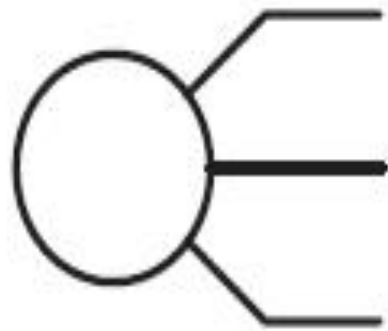


8. MOTOR STARTER

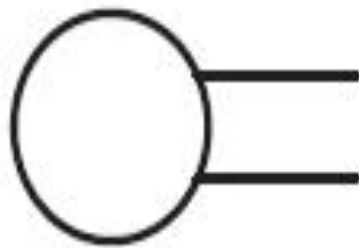


(g) Three-phase magnetic motor starter

9. AC MOTORS



Three-phase
motor



Single-phase
motor



(m) AC motors

10. VARIABLE FREQUENCY DRIVE (VFD)

- A VARIABLE-FREQUENCY DRIVE CONTROLS THE SPEED OF AN AC MOTOR BY VARYING THE FREQUENCY SUPPLIED TO THE MOTOR.
- THE DRIVE ALSO REGULATES THE OUTPUT VOLTAGE IN PROPORTION TO THE OUTPUT FREQUENCY TO PROVIDE A RELATIVELY CONSTANT RATIO (VOLTS PER HERTZ; V/HZ) OF VOLTAGE TO FREQUENCY, AS REQUIRED BY THE CHARACTERISTICS OF THE AC MOTOR TO PRODUCE ADEQUATE TORQUE.

11. VARIABLE FREQUENCY DRIVE (VFD)



NEMA vs IEC

If we compare the NEMA magnetic motor starter to the IEC magnetic motor starter, the following differences would be noticed:

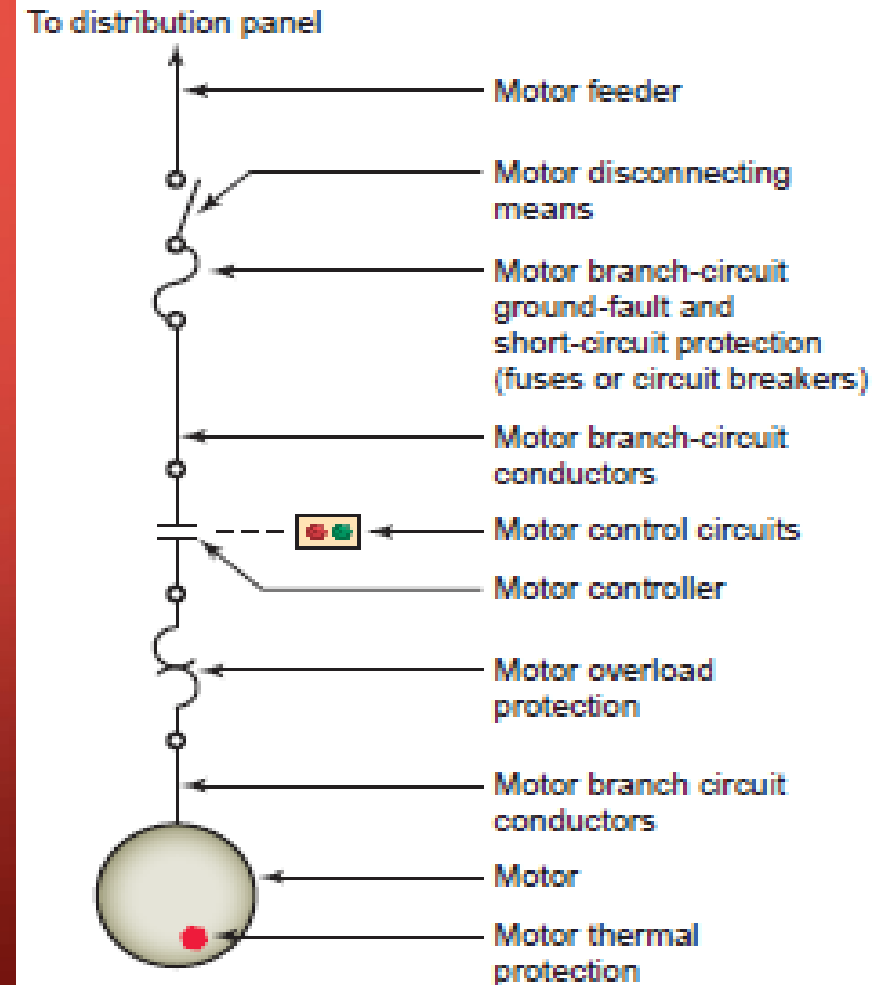
- An IEC device is physically smaller than a comparable NEMA device.
- An IEC device is usually less expensive than a comparable NEMA device.
- An IEC device has a life cycle of approximately one million operations while a comparable NEMA device has a life cycle of almost four times that number.
- An IEC device should normally be protected with fast-acting, current-limiting fuses while a NEMA device can be protected with conventional time delay fuses.

ELECTRICAL CODES AND STANDARDS

NATIONAL ELECTRICAL CODE (NEC)

- Comprises a set of rules that, when properly applied, are intended to provide a safe installation of electrical wiring and equipment. This widely adopted minimum electrical safety standard has as its primary purpose “the practical safeguarding of persons and property from hazards arising from the use of electricity.”
- The single-line diagram illustrates some of the motor terminology used throughout the Code and by motor control equipment manufacturers.

Remember: NEC provided the set of rules we are using in designing and reading ladder diagrams and single-line diagrams.



ELECTRICAL CODES AND STANDARDS

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

- Based in North America, a group that defines and recommends safety standards for electrical equipment. Standards established by NEMA assist users in proper selection of industrial control equipment. As an example, NEMA standards provide practical information concerning the rating, testing, performance, and manufacture of motor control devices such as enclosures, contactors, and starters.

ELECTRICAL CODES AND STANDARDS

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

- The International Electrotechnical Commission (IEC) is a Europe-based organization made up of national committees from over 60 countries.
- Dimensionally, IEC standards are expressed in metric units.
- Like NEMA, IEC provides practical information concerning the rating, testing, performance, and manufacture of motor control devices.

ANSI Symbol	ANSI Code	IEC 617 Symbol	IEC Code	Description
	CON		KM	Contactor contact open
	CON		KM	Contactor contact closed
	CR		KA	Relay contact open
	CR		KA	Relay contact closed
	TR		KT	Timed contact N.O. - ON-delay (TDE)
	TR		KT	Timed contact N.C. - ON-delay (TDE)
	TR		KT	Timed contact N.C. - OFF-delay (TDD)
	TR		KT	Timed contact N.O. - OFF-delay (TDD)
	SS		SA	Selector switch
	PB		SB	Pushbutton N.O.
	PB		SB	Pushbutton N.C.
	PB		SB	Pushbutton mushroom head
	FL		SL	Liquid level switch
	FLS		SF	Flow switch
	PS		SP	Pressure switch
	TS		ST	Temperature switch
	LS		SQ	Limit switch
	PRS		SQ	Proximity switch

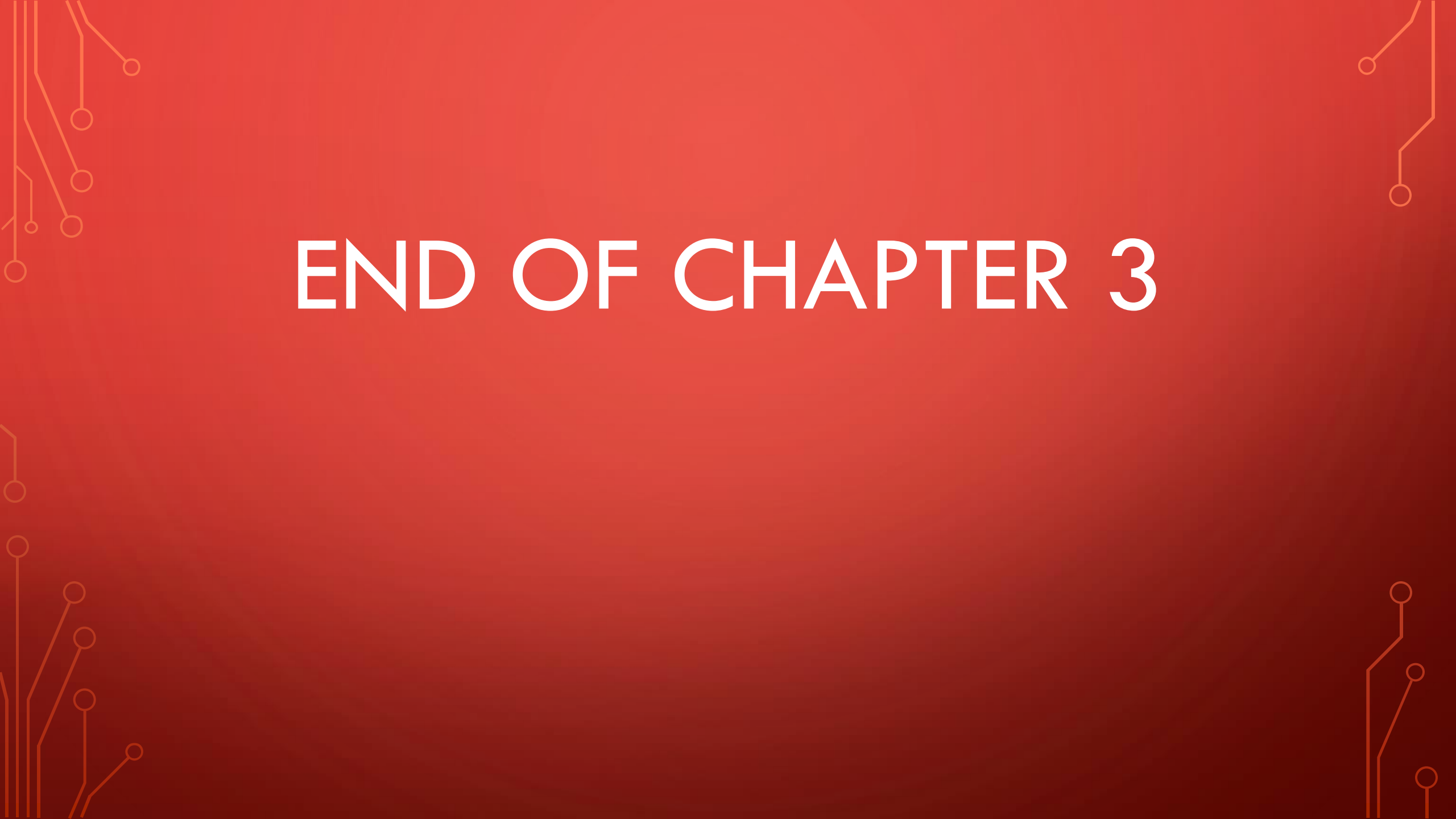
ANSI Symbol	ANSI Code	IEC 617 Symbol	IEC Code	Description
	LT		HL	Indicating light
	PL		XS	Plug and socket
	CR		KA	Control relay coil
	CON		KM	Contactor coil
	M		KM	Motor starter coil
	TR		KA	Timer coil
	SOL		YV	Solenoid coil
	CTR		EC	Electromechanical counter
	CB		QF	Circuit breaker
	T1		X1	Terminals (reference)
	FU		XT	Fused terminals (reference)
	FU		FU	Fuse, protective
				Mechanically-linked (free-guided) contacts
			EStop	Emergency Stop button (mushroom head)
				Positive opening operation of a switch
				Positive opening, latching with twist to reset

*From ANSI/NFPA 79 and ANSI Y 32.2/IEEE 315

COMMON MOTOR CONTROL SYMBOLS

REFERENCE:

ELECTRIC MOTORS AND CONTROL SYSTEM, FRANK D. PETRUZELLA, pg .16

The image features a solid red background with a subtle gradient. In the four corners, there are decorative elements resembling circuit board traces or neural network connections. These elements consist of thin, light-orange lines that branch out and terminate in small circles, creating a symmetrical, geometric pattern.

END OF CHAPTER 3